LIUDONG CHEN

500 W 120th Street, Columbia University, New York, NY, 10027
lc3671@columbia.edu | Personal website | (347) 728-6670

EDUCATION

Columbia University, New York, NY *Sep. 2022 –*
Ph.D. in Earth and Environmental Engineering
Advisor: Prof. Bolun Xu
Tentative thesis: Behavior Modeling and Analysis in Energy Systems

North China Electric Power University, Beijing, China *Sep. 2019 – Jun. 2022*
Master of Science in Electrical Engineering
Advisor: Prof. Nian Liu
Thesis: Multi-Agent Coordinated Optimization for Power Distribution Systems under Uncertainty

North China Electric Power University, Beijing, China *Sep. 2015 – Jun. 2019*
Bachelor of Science in Electrical Engineering and Automation
Exchange student, University of Wisconsin–Milwaukee (one semester)

INDUSTRY EXPERIENCE

Independent System Operator - New England, Holyoke, MA *May. 2025 – Aug. 2025*
Research Intern at Advanced Technology Solution
Mentor: Dr. Feng Zhao

- Proved the conditions under which a proxy price can represent the economic flow direction between two regions in coordinated transaction scheduling (CTS) between regional markets
- Formulated a theoretical game framework demonstrating how hedgers in CTS can shift clearing results across three regions, leading to outcomes that may benefit or hinder CTS operations

National Renewable Energy Laboratory, Golden, CO *May. 2023 – Sep. 2023*
Research Intern at Power Systems Engineering Center
Mentor: Prof. Bai Cui, Prof. Xiangqi Zhu, Dr. Ahmed Zamzam

- Proposed a multistage stochastic optimization model to derive day-ahead multi-segment offer curves for maximizing virtual power plant profits
- Developed a flexibility envelope learning framework to quantify electricity customers' flexibility using low-resolution load data

PUBLICATIONS

Preprints

[P1] **Liudong Chen**, Jay Sethuraman, and Bolun Xu, "Gaming on Coincident Peak Shaving: Equilibrium and Strategic Behavior," *arXiv preprint*:2501.02792, 2025.

[P2] **Liudong Chen** and Bolun Xu, "Equitable Time-Varying Pricing Tariff Design: A Joint Learning and Optimization Approach," *arXiv preprint*:2307.15088, 2023.

[P3] **Liudong Chen**, Bolun Xu, Bai Cui, and Shixuan Zhang, "Optimal Offer Curve Design of a Virtual Power Plant Considering Multi-Stage Uncertainties," *Available at SSRN*:5069035, 2024

Journal Papers

[J1] **Liudong Chen** and Bolun Xu, "A Prudent Framework for Understanding Risk-Awareness in Demand Response," *European Journal of Operational Research*, to be published.

[J2] Jianxiao Wang*, **Liudong Chen***, Zhenfei Tan, Ershun Du, Nian Liu, Jing Ma, Mingyang Sun, Canbing Li, Jie Song, Xi Lu, Chin-Woo Tan, and Guannan He, "Inherent spatiotemporal uncertainty of renewable power in China," *Nature Communications*, 14, 5379, 2023. (* authors contributed equally)

[J3] **Liudong Chen**, Nian Liu, Liangying Liu, Xinghuo Yu, and Yusheng Xue, "Data-driven Stochastic Game with Social Attributes for Peer-to-peer Energy Sharing," *IEEE Transactions on Smart Grid*, vol. 12, no. 6, pp. 5158-5171, 2021.

[J4] **Liudong Chen**, Nian Liu, Songnan Yu, and Yan Xu, "A Stochastic Game Approach for Distributed Voltage Regulation among Autonomous PV Prosumers," *IEEE Transactions on Power System*, vol. 37, no. 1, pp. 776-787, 2022.

[J5] **Liudong Chen**, Nian Liu, and Jianhui Wang, "Peer-to-peer Energy Sharing in Distribution Networks with Multiple Sharing Regions," *IEEE Transactions on Industrial Informatics*, vol. 16, no. 1, pp. 6760-6771, 2020.

[J6] **Liudong Chen**, Nian Liu, Chenchen Li, and Jianhui Wang, "Peer-to-Peer Energy Sharing With Social Attributes: A Stochastic Leader-Follower Game Approach," *IEEE Transactions on Industrial Informatics*, vol. 17, no. 4, pp. 2545-2556, 2021.

[J7] **Liudong Chen**, Nian Liu, Chenchen Li, and Lei Wu, "Multi-Party Stochastic Energy Scheduling for Industrial Integrated Energy Systems Considering Thermal Delay and Thermoelectric Coupling," *Applied Energy*, vol. 304, 117882, 2021.

[J8] **Liudong Chen**, Nian Liu, Chenchen Li, Silu Zhang, and Xiaohe Yan, "Peer-to-peer energy sharing with dynamic network structures," *Applied Energy*, vol. 291, 116831, 2021.

[J9] Nian Liu, Chenchen Li, **Liudong Chen**, and Jianhui Wang, "Hybrid Data-driven and Model-based Distribution Network Reconfiguration with Lossless Model Reduction," *IEEE Transactions on Industrial Informatics*, vol. 18, no. 5, pp. 2943-2954, 2022.

[J10] Yubing Chen, Nian Liu, **Liudong Chen**, and Xinghuo Yu, "Region-to-Region Energy Sharing for Prosumer Clusters in Distribution Network: A Multi-Leaders and Multi-Followers Stackelberg Game," *IEEE Transactions on Energy Markets, Policy and Regulation*, vol. 1, no. 4, pp. 398-409, 2023.

[J11] Chenchen Li, Nian Liu, **Liudong Chen**, Yubing Chen, and João PS Catalão, "Structural Decomposition Approach for Distribution Network Reconfiguration," *IEEE Transactions on Industrial Informatics*, 2025.

Conference Papers

[C1] **Liudong Chen**, Xiangqi Zhu, Bolun Xu, and Fei Ding, "Demand Side Flexibility Envelope Quantification Under Data Scarcity," *IEEE Power & Energy Society Innovative Smart Grid Technologies Conference (ISGT)*, 2024.

[C2] **Liudong Chen** and Bolun Xu, "Saturation Effects in Equitable Demand Response Tariff Design," *North American Power Symposium (NAPS)*, 2023.

Patent

[A1] Nian Liu, **Liudong Chen**, and Chenchen Li, "Method and system of hybrid data-and-model-driven hierarchical network reconfiguration," *US Patent*, 11831505, 2023.

SELECTED PRESENTATIONS

[1] **(Oral & Poster)** "Efficiency Evaluation and Improvement of Coordinated Transaction Scheduling," Independent System Operator - New England Intern Poster Expo, Aug. 2025, Holyoke, U.S.

[2] **(Oral)** "A Prudent Framework for Understanding Risk-Awareness in Demand Response," Independent System Operator - New England Market and Optimization Group Meeting, Aug. 2025, Holyoke, U.S.

[3] **(Poster)** "Gaming on Coincident Peak Shaving: Equilibrium and Strategic Behavior," 2025 LANL Grid Science Winter School and Conference, Jan. 2025, Santa Fe, U.S.

[4] **(Oral)** "Gaming on Coincident Peak Shaving: Equilibrium and Strategic Behavior," 6th Annual Columbia University EEE Graduate Student Symposium, Feb. 2025, New York, U.S.

[5] **(Poster)** "Prudent Price-Responsive Demands," 7th NREL Autonomous Energy Systems Workshop, Sep. 2024, Golden, U.S.

[6] **(Oral)** "Prudent Price-Responsive Demands," ACM SIGEnergy Graduate Student Seminar, May. 2024, Online. Hosted by Dr. Ahmed S. Alahmed.

[7] **(Oral & Poster)** "Equitable Time-Varying Pricing Tariff Design: A Joint Learning and Optimization Approach," 6th NREL Autonomous Energy Systems Workshop, Sep. 2023, Denver, U.S.

[8] **(Oral)** "Equitable Time-Varying Pricing Tariff Design," 2023 North American Power Symposium (NAPS), Oct. 2023, Asheville, U.S.

[9] **(Oral)** "Optimal Offering Strategy of a Price-Taking Virtual Power Plant," 5th Annual Columbia University EEE Graduate Student Symposium, Sep. 2023, New York, U.S.

[10] **(Poster)** "Prudent Price-Responsive Demands," 2024 IEEE PES General Meeting, Jul. 2024, Seattle, U.S.

[11] **(Poster)** "Demand Side Flexibility Envelope Quantification Under Data Scarcity," 2024 IEEE Innovative Smart Grid Technologies, North America (ISGT NA), Feb. 2024, D.C., U.S.

[12] **(Poster)** "Saturation Effects in Equitable Demand Response Tariff Design," 2023 IEEE PES General Meeting, Jul. 2023, Orlando, U.S.

[13] **(Poster)** "Saturation Effects in Equitable Demand Response Tariff Design," 2023 Columbia University Data Science Day, Apr. 2023, New York, U.S.

[14] **(Oral)** "Cyber Security in Interactive Demand Response," IEEE 6th Conference on Energy Internet and Energy System Integration (EI2), Oct. 2021. Taiyuan, China.

TEACHING EXPERIENCE

EEAE4220 Energy System Economics and Optimization, Columbia University *Fall. 2024*
Teaching Assistant. *Instructor*: Prof. Bolun Xu

- Fundamentals of power system economics and optimization algorithms.
- Held weekly office hours, designed and graded assignments, group projects, and exams.

ELEN6906 Future Energy: Economics, Systems, Policies, Columbia University *Spring. 2025*
Teaching Assistant. *Instructor*: Prof. Debasis Mitra

- A holistic view of the impact of climate change on our approach to energy, integrating perspectives from economics, science/engineering, and public policy.
- Held weekly office hours, participated in lecture presentations and discussions, graded assignments,

student presentations, mid-term papers, group projects, and final papers.

MENTORING EXPERIENCE

Research Mentor, Columbia University

- Liwen Peng, M.S. in Data Science, Columbia University *Sep. 2023 – Jun. 2024*
  Research topic: Demographics-driven electrification path in New York City
- Zhongyi Meng, M.S. in Mechanical Engineering, Columbia University *Sep. 2023 – Jan. 2024*
  Research topic: High-granularity thermal load generation modeling for buildings
- Justin Alexander Clarke, B.S. in Computer Science, Columbia University *Jun. 2024 – May. 2025*
  Research topic: Socio-economic dependencies of electrification potential in New York City

Research Mentor, North China Electric Power University (NCEPU)

- Yufan Lu, Ph.D. in Electrical Engineering, NCEPU *Sep. 2020 – Jun. 2022*
  Research topic: Emergency resources scheduling and outage recovery for power distribution system.
- Yubing Chen, M.S. in Electrical Engineering, NCEPU *Sep. 2020 – Jun. 2022*
  Research topic: Region-to-region energy sharing for prosumer clusters in power distribution systems
- Liangying Liu, M.S. in Electrical Engineering, NCEPU *Sep. 2020 – Jun. 2021*
  Research topic: Modeling prosumers' demand-shifting preferences in energy sharing based on survey data and load metering data

SERVICES & ACTIVITIES

Journal Reviewer: IEEE Transactions on Smart Grid; IEEE Transactions on Power System; IEEE Transactions on Industrial Informatics; IEEE Transactions on Sustainable Energy; IEEE Transactions on Energy Markets, Policy and Regulation; Applied Energy; Energy; Renewable Energy; Sustainable Energy, Grids and Networks, etc.

Conference Reviewer: 2025 IEEE PES General Meeting; 12th Bulk Power System Dynamics and Control Symposium (IREP'2025); 2023 North American Power Symposium (NAPS), etc.

Conference Volunteer: 2023 IEEE PES General Meeting; 2018 IEEE Conference on Energy Internet and Energy System Integration (EI2); 2019 and 2020 IEEE Women in Engineering (WIE) Beijing.

Other Services: Vice president of NCEPU IEEE student chapter (2021). Supervised preliminary research work for high school students in New York. Organized and participated in numerous seminars to help junior undergraduate and graduate students in NCEPU better plan their academic and professional paths.

HONORS & AWARDS

- Outstanding Student Scholarship, IEEE Power & Energy Society *2022*
- Grid Science Winter School Scholarship, Los Alamos National Lab *2025*
- Autonomous Energy System Workshop Scholarship, National Renewable Energy Laboratory *2024*
- National Scholarship, Chinese Ministry of Education (top 0.5%) *2018, 2020, 2021*
- President Scholarship, North China Electric Power University (top 0.1%) *2021*
- Ultra-high Voltage Scholarship, State Grid Corporation of China (top 0.3%) *2018*
- Excellent Postgraduate Students' Award, Department of Education of Beijing *2022*
- Excellent graduate Students' Award, Department of Education of Beijing *2019*
- National First Prize, China Postgraduate Mathematical Contest in Modeling (top 1.5%) *2019*
- Meritorious Winner, Mathematical & Interdisciplinary Contest In Modeling (MCM&ICM) *2018*